Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

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Abstract

We explore how generating a chain of thought, a series of intermediate reasoning steps, significantly improves the ability of large language models to perform complex reasoning. In particular, we show how such reasoning abilities emerge naturally in sufficiently large language models via a simple method called chain-of-thought prompting, where a few chain of thought demonstrations are provided as exemplars in prompting. Experiments on arithmetic, commonsense, and symbolic reasoning benchmarks show that chain-of-thought prompting improves performance substantially. For instance, on the GSM8K benchmark of math word problems, chain-of-thought prompting with PaLM 540B achieves 56.9 percent solve rate, compared with 17.9 percent for standard prompting. The gains are an emergent property of model scale: chain-of-thought prompting does not help small models, but it produces large gains for models with roughly 100 billion parameters or more.

Keywords: chain-of-thought, prompting, reasoning, large language models, few-shot

1 Introduction

Large language models have achieved impressive performance on many natural language tasks, but they still struggle with tasks that require multi-step reasoning, such as math word problems, commonsense reasoning, and symbolic manipulation. A common approach is to fine-tune the model on a large corpus of reasoning traces, but this requires substantial annotated data and does not generalize well to new tasks.

In this paper we show that a simple prompting strategy, chain-of-thought prompting, elicits reasoning in large language models without any fine-tuning. Instead of asking the model to produce only the final answer, we provide a few exemplars in which the answer is preceded by a series of intermediate reasoning steps. The model then produces a similar chain of thought for a new question. This simple change produces large gains on arithmetic, commonsense, and symbolic reasoning benchmarks, and the gains grow with model scale. Figure 1 shows an example prompt and the generated chain of thought.

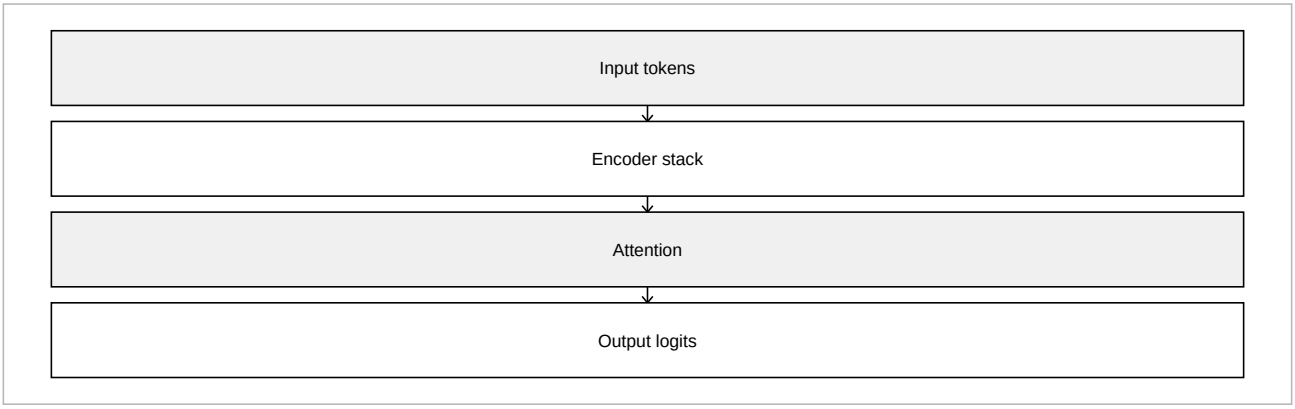

Figure 1: An example of chain-of-thought prompting: the exemplar includes intermediate reasoning steps, and the model generates a similar chain for the test question.

2 Chain-of-Thought Prompting

Chain-of-thought prompting is a few-shot prompting method. For each task we hand-write a small number of exemplars, each consisting of a question, a chain of thought, and an answer. At inference time the exemplars are prepended to the test question, and the model generates a chain of thought followed by an answer. The model is

not fine-tuned; all of its parameters remain frozen.

2.1 Method

We use 8 exemplars for each task. The exemplars are written in natural language and contain the intermediate calculation or reasoning steps that lead to the answer. For arithmetic tasks the chain of thought is a sequence of equations and intermediate results; for commonsense tasks it is a short natural language explanation; for symbolic tasks it is a sequence of string manipulations. The model's output is parsed by taking the last line as the answer, and the answer is compared with the gold answer after normalization.

The key difference from standard prompting is that the exemplars include the reasoning process, not only the final answer. This gives the model a template for decomposing a problem into intermediate steps, and it makes the model's computation more transparent. The prompt is a concatenation of the exemplars and the test question, and Equation (1) shows the answer being selected by the model.

$$\text{answer} = \arg\max_{a} p(a \mid \text{question}, \text{exemplars}) \tag{1}$$

2.2 Comparison with Standard Prompting

Standard prompting uses the same number of exemplars but includes only the question and the answer, without the intermediate reasoning. We compare the two methods on the same models and the same tasks. Standard prompting performs well on tasks that require a single step, but it degrades sharply as the number of reasoning steps increases. Chain-of-thought prompting, by contrast, maintains high accuracy on problems that require several steps.

3 Experiments

We evaluate chain-of-thought prompting on three families of reasoning tasks: arithmetic, commonsense, and symbolic reasoning. We use the PaLM family of models, with sizes ranging from 8 billion to 540 billion parameters, as well as GPT-3 and LaMDA. All models are evaluated in the few-shot setting without fine-tuning.

3.1 Arithmetic Reasoning

The arithmetic benchmarks are GSM8K, SVAMP, ASDiv, AQuA, and MAWPS. GSM8K is the most difficult, because its problems require two to eight steps of reasoning. With PaLM 540B, chain-of-thought prompting achieves 56.9 percent solve rate on GSM8K, compared with 17.9 percent for standard prompting. On SVAMP, the same model reaches 79.0 percent with chain-of-thought prompting, compared with 57.4 percent for standard prompting.

The gains are much smaller for models with fewer than 100 billion parameters. For example, PaLM 8B improves only marginally, and GPT-3 175B improves by about 10 points. This suggests that chain-of-thought reasoning is an emergent ability of scale, rather than a generic benefit of the prompt format.

3.2 Commonsense Reasoning

We evaluate on CSQA, StrategyQA, and Date Understanding. Chain-of-thought prompting improves PaLM 540B from 74.5 to 79.2 on CSQA, and from 64.6 to 72.3 on StrategyQA. The improvements are smaller than on arithmetic tasks but consistent across datasets. Many of the generated chains of thought contain plausible commonsense facts, which suggests that the model is using the reasoning trace to retrieve and combine knowledge.

3.3 Symbolic Reasoning

The symbolic tasks are last-letter concatenation, coin flip, and reverse string. These tasks are synthetic and require no world knowledge, which makes them a clean test of multi-step reasoning. Chain-of-thought prompting substantially improves PaLM 540B on last-letter concatenation and coin flip, reaching 91.4 percent and 94.2 percent respectively. Standard prompting remains near chance on the same tasks, which shows that the benefit is not due to memorized answers.

4 Discussion

Why does chain-of-thought prompting work? One explanation is that the generated intermediate steps provide additional computation, effectively allowing the model to use more layers of computation before producing an answer. Another explanation is that the intermediate steps decompose a hard problem into easier sub-problems, each of which the model can solve. Both explanations are consistent with the observation that the gains grow with model scale.

4.1 When Does It Help?

Chain-of-thought prompting helps most when the task requires multi-step reasoning and the model is large enough. It does not help on tasks that require a single lookup, and it can hurt on tasks where the model already performs near ceiling. The quality of the exemplars matters: exemplars with clear and correct reasoning produce better chains than exemplars with noisy reasoning.

4.2 Limitations

The generated chains of thought are not guaranteed to be faithful explanations of the model's computation. The model can produce a correct answer with an incorrect chain, and it can produce a plausible chain that leads to a wrong answer. Chain-of-thought prompting also increases inference cost, because the model generates many more tokens. Finally, the method requires hand-written exemplars, which limits its applicability to new tasks.

5 Related Work

Few-shot prompting was popularized by GPT-3, which showed that large language models can perform many tasks from a handful of examples. Subsequent work has explored prompting strategies such as scratchpads, which also elicit intermediate computation. Our work differs in that it studies the emergence of reasoning as a function of model scale and provides a systematic evaluation across three families of reasoning tasks.

Fine-tuning on reasoning traces has also been shown to improve performance, but it requires annotated data. Chain-of-thought prompting achieves similar gains without any training, which makes it much cheaper to apply to new tasks.

6 Conclusion

We have shown that chain-of-thought prompting, a simple few-shot prompting method, elicits multi-step reasoning in large language models. On arithmetic, commonsense, and symbolic reasoning tasks, the method produces large gains for models with roughly 100 billion parameters or more. We hope that this work encourages further study of how reasoning abilities emerge with scale, and of how generated reasoning traces can be used to verify model outputs.

Table 1: Solve rate on GSM8K with standard prompting and chain-of-thought prompting.

Model	Standard	Chain-of-thought
PaLM 8B	4.1	5.2
PaLM 62B	9.6	18.8
PaLM 540B	17.9	56.9
GPT-3 175B	15.6	26.4

Table 2: Accuracy on commonsense and symbolic reasoning benchmarks with PaLM 540B.

Benchmark	Standard	Chain-of-thought
CSQA	74.5	79.2
StrategyQA	64.6	72.3
Last-letter concatenation	8.2	91.4
Coin flip	49.8	94.2

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A Appendix: Prompts

Each prompt consists of 8 exemplars followed by the test question. For GSM8K the exemplars are taken from the training set and include a chain of thought with the calculation steps. For commonsense tasks the exemplars include a short explanation, and for symbolic tasks they include the intermediate string operations. The prompts are identical across model sizes, and no prompt tuning is performed.

We report results with greedy decoding and with temperature 0. The answer is extracted from the last line of the generated text, and we normalize numbers and strings before comparison. All benchmarks use their standard test splits, and we do not train on any of the evaluation data.